Target Detection With Spectral Graph Contrast Clustering Assignment and Spectral Graph Transformer in Hyperspectral Imagery

Xi Chen, Maojun Zhang, and Yu Liu

***Abstract*— Hyperspectral target detection (HTD) is a method that recognizes objects of interest in a scene by a priori target spectrum. Local details and global information on the spectra are critical for accurate target identification. Detectors with excellent discrimination of spectral differences can better highlight targets while suppressing background. To this end, this article proposes an HTD method based on spectral graph contrast clustering assignment and the spectral graph transformer (SGT) to solve these problems. Specifically, for local–global feature extraction of spectra, the pixel spectra are first constructed as the spectral graph. Then, the representations of the first- or higher-order neighbors of the nodes in the spectral graph are aggregated using graph convolutional networks to extract the local detail information of the spectra. The self-attention in Transformer is utilized to learn the global information of the spectra. Second, a novel spectral graph contrast clustering assignment method is proposed to equip the model with excellent spectral discrimination ability. It maintains clustering consistency by swapping predictive clustering assignments while maximizing the similarity of semantically similar graph clusters and keeping other semantically different graph clusters away from them to better discriminate differences between spectra. Finally, comparisons with seven state-of-the-art HTD methods on four real hyperspectral datasets and ablation studies verify the effectiveness of the proposed method in HTD.**

***Index Terms*— Contrast clustering, hyperspectral imagery (HSI), spectral graph transformer (SGT), target detection.**

I. Introduction

HYPERSPECTRAL remote sensing is a cutting-edge technological tool in remote sensing science, capable of both imaging and spectroscopy, which is a significant breakthrough in the development of remote sensing science and technology [1]. Each pixel in the hyperspectral imagery (HSI) corresponds to a near-continuous spectral curve, which can reflect the diagnostic spectral absorption differences of substances, providing rich spectral information to extract ground feature information accurately [2]. Hyperspectral target detection (HTD) has received widespread attention as one of the advantageous applications of HSIs and has been widely used in mineral exploration [3], camouflage identification [4], medical diagnosis [5], and environmental monitoring [6].

HSIs have rich spectral information. Unlike traditional target detection methods that use spatial texture and morphological information of ground objects for target recognition, HTD uses the spectral characteristics of objects of interest to identify targets accurately [7]. HTD focuses on the targets of interest in the scene to be detected. It considers all other scene components to be detected except the object of interest as background [8]. However, for HTD, the prior knowledge generally possessed is only the spectral features of the target of interest, and there is no information about the class labels in the scene to be detected. Moreover, phenomena such as the complexity of the background [9] (the background may contain multiple substance classes) and the inherent variation of the spectra of the same substance [10] pose difficulties for HTD. The main challenge of HTD is to accurately identify and localize targets from complex backgrounds based on a priori target spectrum and to be able to suppress the background effectively.

To utilize the spectral characteristics of materials to detect targets of interest in HSIs [11], many HTD methods have been developed in the past decades. Constrained energy minimization (CEM) [12] is the classical matched-filter-based HTD method, which detects the target and suppresses the background by minimizing the overall energy output of a finite impulse response filter within the spectral constraints of the target. Adaptive coherent\cosine estimator (ACE) [13] is a probabilistic statistics-based HTD method that assumes that the background conforms to a multivariate Gaussian distribution and detects the target adaptively from the background based on the covariance matrix of the HSI. Orthogonal subspace projection (OSP) [14], [15] is an HTD method based on subspace modeling, which detects targets by projecting the pixel spectra into orthogonal subspaces of individual background end-elements to suppress the interference of the background. These methods require elaborate artificial feature extraction methods and prior assumptions, and it is challenging to utilize the nonlinear features of spectra, which makes them less adaptable to complex detection scenes. The above HTD methods are extended to corresponding kernel-based nonlinear versions to utilize the nonlinear relationship between spectral bands such as kernel-CEM (KCEM) [16], kernel adaptive coherence estimation (KACE) [17], and kernel OSP

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(KOSP) [18]. Kernel-based HTD methods implicitly map the data into a high-dimensional kernel feature space [19] by using the corresponding kernel function to make the data well separated. However, kernel-based HTD methods rely on the assumption that the transformation of the data into a high-dimensional kernel feature space becomes linearly separable. Representation-based HTD methods have emerged to avoid making any explicit assumptions about the statistical distribution of the data [20] such as the sparse representation-based target detector (STD) [21], combined sparse and co-sparse representation-based HTD method (CSCR) [22], and decomposition model for HTD based on background dictionary learning (DM-BDL) [23]. Although the representation-based HTD methods can achieve good detection results, obtaining a pure background dictionary is difficult due to factors such as noise, and the optimal number of dictionary atoms in the matrix decomposition may differ for different data [24]. It requires human experience to be set, significantly limiting its ability to adapt to different scenarios. Furthermore, to further enhance the performance of HTD, researchers have proposed detectors based on graph theory, such as the HTD method based on sparse representations and Cauchy distance combination graph [25], the target detector based on spectral perception and spatiotemporal tensor decomposition [26], and the target detector based on weighted Cauchy distance graph and local adaptive collaborative representation [27]. These detectors utilize the powerful analytical tool of graph theory to provide new perspectives for detecting and understanding targets and backgrounds in HSIs.

In recent years, thanks to the excellent nonlinear feature representation and advanced semantic feature extraction capabilities of deep neural networks, they have attracted extensive attention from researchers in the fields of hyperspectral classification [28], [29], [30], unmixing [31], super-resolution [32], and change detection [33], and have provided new directions for HTD [34]. However, for HTD, the extreme scarcity of prior information creates difficulties for deep learning-based HTD methods. Some researchers have approached the problem from the perspective of transfer learning in the hope of overcoming the inability to train detectors in a supervised learning manner due to the scarcity of prior information, such as the convolutional neural network-based HTD (CNND) [35] method, the meta-learning and Siamese network-based HTD (MLSN) method [36], the sensor-independent HTD (SIHTD) method [37], and the transfer-learning-based HSI spectral–spatial joint target detection methods [38]. However, using the idea of transfer learning to transfer the model knowledge learned on a dataset with known label information in the source domain to the target domain to be detected scene will be limited by the adaptability of the model knowledge to the detection scene, resulting in a poor target detection performance. In addition to employing transfer learning, some researchers have helped train models by synthetically expanding target and background samples. For example, in a deep convolutional neural network for HTD (HTD-Net) [39], HTD-Net uses a modified self-encoder (U-AE) with contraction and symmetric expansion paths to generate the target samples and then finds the background samples that are significantly different from the target samples based on a linear prediction strategy and then pairs the obtained target with the background samples to enable the model to learn the differences between the target samples and the background samples. An HTD method with an auxiliary generative adversarial network [40] uses a generative adversarial network to generate simulated target and background spectra to augment the target and background samples to train a binary classification HTD-Net. A two-stream CNND method (TSCNTD) [41] trains a binary classification HTD-Net by finding sufficient background samples in the HSI to be detected through a hybrid sparse representation and classification-based pixel selection strategy and then synthesizing the target prior with the typical background samples to obtain sufficient target samples. A Siamese transformer network (STTD) [42] for HTD extracts high-purity background pixels in the HSI to be detected by an endmember extraction and unmixing algorithm and then mixes the target prior with the high-purity background pixels to construct the target and background samples for training a binary classification HTD network. A spectral aggregation and separation network (SASN) with target band random masking (TBRM) for HTD [43] proposes a multivariate representative background selection strategy to obtain the representative background samples and generates sufficient target samples by repeating the randomized zero-masking prior target spectral bands to train the detector. Although these deep-learning-based detectors achieve excellent detection performance, they still need to be improved by the quality of the expanded target and background samples, which may introduce knowledge bias into the model.

To overcome the above problems, we plan to construct a spectral discrimination pretext task with self-supervised learning to fully exploit the information of the HSI itself so that the model can discriminate the spectral differences and then eliminate the dependence on the explicit target and background training samples. In addition, we observe that the local detail information and global information of the spectrum are essential for accurate target identification. Therefore, in this article, we propose a spectral graph contrast clustering assignment for HTD using a spectral graph transformer (SGT) to address these issues, denoted as SGCCA. SGCCA transforms spectra into spectral graph data, extracts the local detail information of spectra by aggregating the representations of the first- or higher-order neighbors of the nodes in the spectral graph via graph convolution network (GCN), and then obtains the global features of the spectra by using self-attention in Transformer to obtain the local–global features of the spectra. In addition, we propose a spectral graph contrast clustering assignment method for spectral discriminative learning, which helps the model to better discriminate the differences and similarities between spectra by swapping the predictive clustering assignments to maintain the clustering consistency while maximizing the similarity of two semantically similar graph clusters and keeping other semantically different graph clusters away from them.

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The main contributions of this article are summarized as follows.

1) The pixel spectra are constructed as spectral graphs, and the structural characteristics of the spectral graphs are utilized to obtain the local–global spectral features of the spectra through the designed SGT.
2) A spectral discriminative learning method by spectral graph contrast clustering assignment is proposed to obtain a model with spectral discriminative ability in a self-supervised learning manner, free from the dependence on displaying supervised information.

The remainder of this article is organized as follows. Section II briefly describes the related work covered in this article. In Section III, we describe the principle of the proposed method in detail. Then, we verify the effectiveness and superiority of the proposed method through extensive experiments and analysis in Section IV. Finally, we summarize the work of this article in Section V.

II. Related Work

A. Graph Convolution Networks

Let $G = (V, E)$ denote an undirected graph, where $V = \{v_i\}_{i=1,\ldots,N}$ denotes the set of nodes with N nodes and $E = \{e_{ij}\}_{i,j=1,\ldots,N}$ denotes the set of edges between nod. The adjacency matrix of G is $\mathbf{A} \in \mathbb{R}^{N\times N}$, and the degree matrix is $\mathbf{D} = \Sigma_j \mathbf{A}_{ij}$. The feature matrix of nodes is $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N\}$, where $\mathbf{x}_i \in \mathbb{R}^{d_o}$ is the node feature of node v_i. In general, modern graph neural networks follow the learning model of iteratively updating the representations of nodes by aggregating the representations of their first- or higher-order neighbors [44]. For graph convolutional networks [45], the aggregation operation on graph nodes can be formalized as

$$\mathbf{X}^{(l+1)} = \sigma\left(\tilde{\mathbf{D}}^{-\frac{1}{2}}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-\frac{1}{2}}\mathbf{X}^{(l)}\mathbf{W}^{(l)}\right) \tag{1}$$

where σ denotes the nonlinear activation function, $\tilde{\mathbf{D}}_{ii} = \Sigma_j \tilde{\mathbf{A}}_{ij}$ is the degree matrix of the undirected graph G with self-connections, $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ is the adjacency matrix of the undirected graph G with self-connections, $\mathbf{I}$ is the identity matrix, and $\mathbf{W}^{(l)}$ is the trainable weight matrix.

B. Transformer Encoder

The Transformer encoder consists of a multihead self-attention module with a feedforward network. The feedforward network has two linear transformations with a ReLU activation function in the middle. Multihead self-attention is realized by running h self-attention operations called "heads" in parallel and projecting their cascaded outputs [46], as shown in Fig. 1.

For the input sequence of node features $\mathbf{z} \in \mathbb{R}^{N\times d}$, the query $\mathbf{Q}_i$, key $\mathbf{K}_i$, and value $\mathbf{V}_i$ are obtained by the projection of the learnable feature matrix. The process can be represented as

$$[\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i] = \mathbf{z}\mathbf{E}_{Q_iK_iV_i}, \quad i = 1, \ldots, h \tag{2}$$

where $\mathbf{E}_{Q_iK_iV_i} \in \mathbb{R}^{d\times 3d_k}$ is the learnable projection matrix to project the input sequence into query $\mathbf{Q}_i$, key $\mathbf{K}_i$, and value $\mathbf{V}_i$.

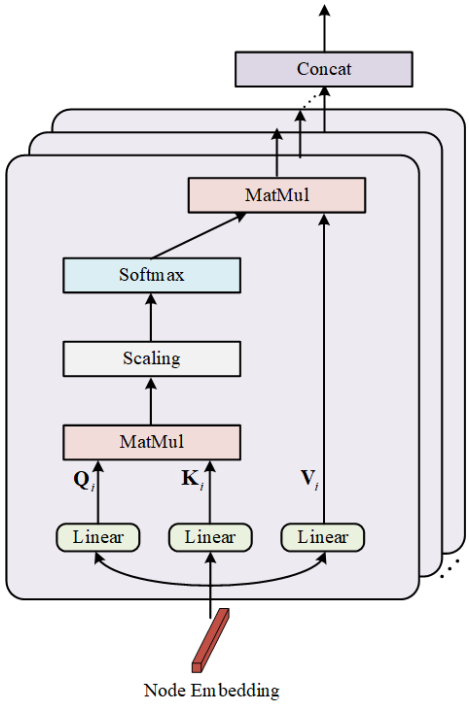


Fig. 1. Multihead self-attention module.

For each head, the dot product of the query $\mathbf{Q}_i$ with the key $\mathbf{K}_i$ is computed, and the resulting dot product value is scaled by dividing it by $\sqrt{d_k}$. The weight of the value $\mathbf{V}_i$ is then obtained by the softmax function, and then the weight is multiplied by the value $\mathbf{V}_i$ to obtain the self-attention of the embedded node feature sequence [47]. The process can be formalized as

$$\mathrm{SA}_i(\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i) = \mathrm{soft\,max}\left(\frac{\mathbf{Q}_i\mathbf{K}_i^{\mathrm{T}}}{\sqrt{d_k}}\right)\mathbf{V}_i. \tag{3}$$

Finally, projecting their cascaded outputs to obtain the output of the multihead self-attention module can be expressed as

$$\mathrm{MSA}(\mathbf{z}) = [\mathrm{SA}_1(\mathbf{Q}_1, \mathbf{K}_1, \mathbf{V}_1); \mathrm{SA}_2(\mathbf{Q}_2, \mathbf{K}_2, \mathbf{V}_2); \ldots; \mathrm{SA}_h(\mathbf{Q}_h, \mathbf{K}_h, \mathbf{V}_h)]\mathbf{E}_{\mathrm{MSA}} \tag{4}$$

where $\mathbf{E}_{\mathrm{MSA}} \in \mathbb{R}^{h\cdot d_k\times d}$ is the feature matrix of the output of the projection cascade, and h is the number of self-attention operations used in parallel. To make the total computational cost of multihead attention similar to that of full-dimensional single-head attention, d_k is generally set to d/h.

C. Self-Supervised Learning

Self-supervised learning belongs to unsupervised learning, which mainly mines its supervisory information from unlabeled data, and valuable representations can be learned by training the model with this constructed effective supervisory information. Contrastive learning is a method to realize self-supervised learning, widely used in traditional computer vision tasks. Such as a simple framework for contrastive learning of visual representations (SimCLR) [48], unsupervised learning of visual features by online contrast clustering assignment

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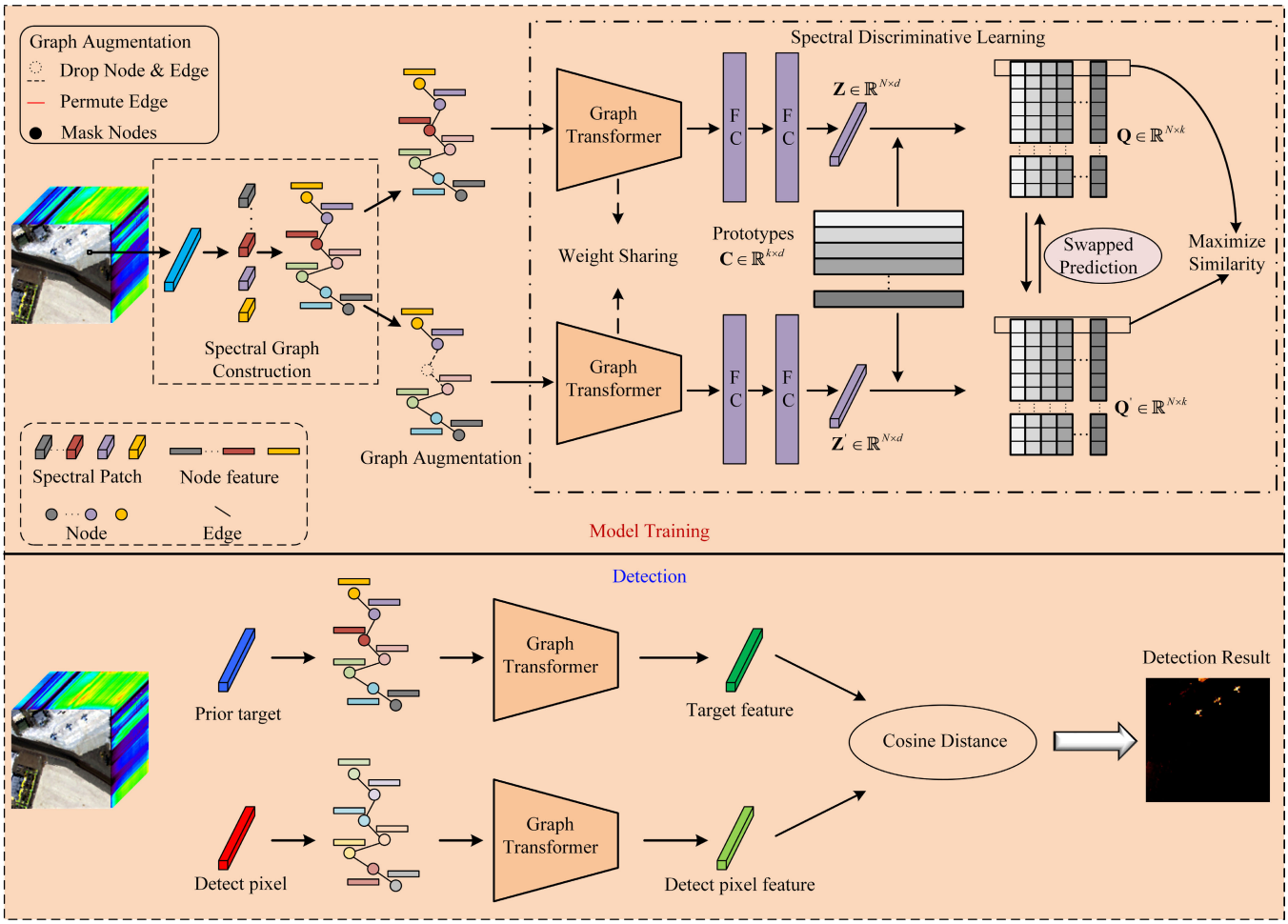


Fig. 2. Framework of SGCCA. SGCCA consists of model training and target detection, whereas model training includes three parts: SGC, graph augmentation, and spectral discriminative learning.

(SwAV) [49], momentum contrastive learning (MoCo) [50] for unsupervised visual representations and BYOL, a new method for self-supervised image representation learning without negative sample pairs proposed in [51]. SimCLR directly uses negative samples coexisting in the current batch and requires a large batch size to work properly. Moco maintains a queue of negative samples and turns a branch into a momentum encoder to improve the consistency of the queue [52].

In recent years, self-supervised learning has demonstrated significant potential in HSI processing, where it has been widely applied in applications such as HTD, hyperspectral anomaly detection (HAD), and HSI denoising. For instance, the self-supervised spectral-level contrastive learning for HTD (SCLHTD) [53] method, based on self-supervised spectral-level contrastive learning, enables the model to discern spectral differences and achieve HTD. The new blind-spot self-supervised learning network for HAD [54] trains the network in a self-supervised way, producing lower reconstruction errors for the background and weakening feature expression for anomalies. The self-supervised deep-learning network for HSI denoising [55] employs the self-supervised learning technique where the model trains itself to learn one part of the noisy input from another part of the noisy input. The ability of self-supervised learning to extract robust and discriminative features, even in the absence of labeled data, highlights its noteworthy potential for future HSI-processing tasks.

III. Proposed Method

This section describes the proposed method in detail. Fig. 2 illustrates the method block diagram of the SGCCA method, which consists of three main parts: spectral graph construction (SGC) and graph augmentation, spectral discriminative learning, and target detection. SGCCA first constructs the pixel spectra into the spectral graphs with the expectation of utilizing the structure and characteristics of the graphs so that the designed SGT can extract more discriminative spectral local–global features from them. The constructed spectral graphs are then subjected to data augmentation, by which training samples for spectral discriminative learning can be constructed. The original spectral graph and its corresponding data-augmented spectral graph should be clustered into the same cluster in the clustering space, which is the key that the model can be used without having to mine the explicit target and background samples. Spectral discriminative learning allows the model to discriminate spectral differences by swapping the clustering assignments of the predicted spectral graphs and maintaining their clustering consistency while maximizing the similarity of semantically similar clusters and

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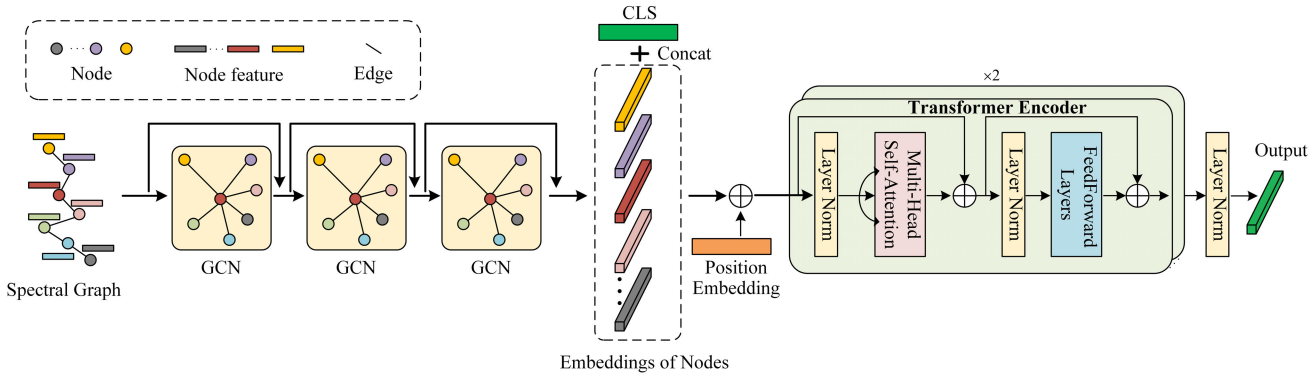


Fig. 3. Schematic of the SGT. The SGT consists of three layers of GCNs and two layers of Transformer encoders. GCNs are employed to capture local details in the spectrum, while Transformer encoders are used to focus on the global information of the spectrum.

keeping other semantically different clusters away from them. Target detection is achieved by extracting the representation of the prior target spectrum and the spectra of the pixels to be detected from the trained SGT with spectral discrimination capability, respectively, and obtaining the target detection result based on the cosine distance between the representation of the spectra of the pixels to be detected and the representation of the prior target spectrum.

A. SGC and Graph Augmentation

1) Spectral Graph Construction: Each pixel spectrum $\mathbf{x}_i \in \mathbb{R}^{1\times B}$ is divided into n spectral blocks $\{\mathbf{x}_i^1, \mathbf{x}_i^2, \ldots \mathbf{x}_i^n\}$ with spectral features $\mathbf{x}_i^n \in \mathbb{R}^{1\times(B/n)}$ for the original HSI $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_{H\times W}] \in \mathbb{R}^{(H\times W)\times B}$. The n spectral blocks $\{\mathbf{x}_i^1, \mathbf{x}_i^2, \ldots \mathbf{x}_i^n\}$ are regarded as n nodes in the spectral graph, and the features of the nodes are the corresponding spectral block features. If two spectral blocks are adjacent to each other in the original pixel spectra, the connecting edges between the corresponding nodes on the spectral graph can be expressed as

$$\mathbf{A}_{i,j} = \begin{cases} 1, & \text{if } \mathbf{x}_i^i \in N_1(\mathbf{x}_i^j) \text{ or } \mathbf{x}_i^j \in N_1(\mathbf{x}_i^i) \\ 0, & \text{otherwise} \end{cases} \tag{5}$$

where $N_1(\mathbf{x}_i^j)$ is the set of *1*-hop neighbors of node $\mathbf{x}_i^j$ [56].

2) Spectral Graph Data Augmentation: The purpose of SGDA is to create novel, realistic, and reasonable spectral graph augmentation data by SGDA without affecting the semantics of spectral graph [57]. A batch of spectral graph data $G \in \{G_1, G_2, \ldots, G_M\}$ has M spectral graphs. Node dropping, edge perturbation, and masking nodes are performed randomly for each spectral graph. For a given spectral graph G_i, node dropping will randomly discard some nodes and the corresponding connected edges. Edge perturbation perturbs the structure of the spectral graph by randomly adding or dropping a specific ratio of edges. Mask nodes replace a certain proportion of node features with values sampled from a random normal distribution. After SGDA, the corresponding augmented spectral graph $G' \in \{G'_1, G'_2, \ldots, G'_M\}$ for this batch can be obtained, G_i and G'_i can be regarded as a pair of positive samples. Using the augmented spectral graph to train the model not only enhances the generalization ability of the model but also better enables the model to capture the essential semantic information in the spectral graph.

B. Spectral Graph Transformer

The spectral graph representation is extracted in SGCCA using SGT; the specific structure of the SGT is shown in Fig. 3. The SGT consists of a three-layer GCN and a two-layer Transformer encoder. The GCN iteratively updates the representations of the nodes by aggregating the representations of the first- or higher-order neighbors of the nodes. Since the features of the nodes in the spectral graph are the spectral blocks in the original spectra, iteratively updating the representations of the nodes by aggregating the first- or higher-order neighbors of the nodes through the graph convolution captures the local details of the spectral. Before feeding the node embeddings obtained after three layers of GCN into the Transformer encoder, a learnable CLS marker is appended to the node embeddings, and position encoding is added. The output embedding corresponding to the position obtained after the CLS marker passes through the Transformer encoder serves as the global representation of the spectral graph [58]. The position encoding uses the different frequencies of the sine and cosine functions [47], that is,

$$\begin{aligned} \mathrm{PE}_{(\mathrm{pos},2i)} &= \sin\left(\frac{\mathrm{pos}}{10000^{2i/d_{\mathrm{GCN}}}}\right) \\ \mathrm{PE}_{(\mathrm{pos},2i+1)} &= \cos\left(\frac{\mathrm{pos}}{10000^{2i/d_{\mathrm{GCN}}}}\right) \end{aligned} \tag{6}$$

where pos is the position, i denotes the number of dimensions, and d_{GCN} is the dimension of the node embedding after three layers of the GCN. Benefiting from the self-attention mechanism in the Transformer encoder, the CLS markers can interact with all node embeddings, and thus obtain global information about the spectrum.

After SGC with SGDA, $G \in \{G_1, G_2, \ldots, G_M\}$ and $G' \in \{G'_1, G'_2, \ldots, G'_M\}$ are fed into the SGT to extract their representations, and the process can be represented as

$$\begin{aligned} Y &= f_\theta(G) \\ Y' &= f_\theta(G') \end{aligned} \tag{7}$$

where $f_\theta(\cdot)$ denotes the SGT, θ is the learnable parameter in the SGT, $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_M] \in \mathbb{R}^{M\times d_{\mathrm{GT}}}$ is

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the output of the original spectral graph through the SGT, $\mathbf{Y}' = [\mathbf{y}'_1, \mathbf{y}'_2, \ldots, \mathbf{y}'_M] \in \mathbb{R}^{M \times d_{\mathrm{GT}}}$ is the output of the spectral graph augmented samples through the SGT, and d_{GT} is the feature dimensions of the SGT output.

C. Spectral Discriminability Learning

To train models with the ability to discriminate spectral differences without relying on displaying target and background samples, a spectral discriminative learning method for spectral graph contrast clustering assignment is proposed to learn spectral discriminative ability in an unsupervised manner. Spectral graph contrast clustering assignment maintains clustering consistency [59] by swapping predictive clustering assignments while maximizing the similarity of semantically similar spectral graph clustering clusters and keeping her semantic spectral graph clustering clusters away from them to better discriminate differences between spectra.

Formally, for a batch size M spectral graph $G \in \{G_1, G_2, \ldots, G_M\}$ with spectral graph data augmentation (SGDA) samples $G' \in \{G'_1, G'_2, \ldots, G'_M\}$. G_i and G'_i are fed into a parameter-sharing SGT with a projection head to extract their features to obtain $\mathbf{Z} = [\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_M] \in \mathbb{R}^{M \times d}$ and $\mathbf{Z}' = [\mathbf{z}'_1, \mathbf{z}'_2, \ldots, \mathbf{z}'_M] \in \mathbb{R}^{M \times d}$, where d is the dimension the projection head output. Since a batch of spectral graphs may possess multiple spectral graphs belonging to the same semantics, the spectral graphs with their augmented samples of features $\mathbf{Z}$ and $\mathbf{Z}'$ can be clustered into k clusters, with the centroid of each cluster represented by a set of learnable prototypes $\mathbf{P} = [\mathbf{p}_1, \mathbf{p}_2, \ldots, \mathbf{p}_k] \in \mathbb{R}^{k \times d}$.

After cluster assignment, the cluster similarities $\mathbf{Q} = [\mathbf{q}_1, \mathbf{q}_2, \ldots, \mathbf{q}_M] \in \mathbb{R}^{M \times k}$ and $\mathbf{Q}' = [\mathbf{q}'_1, \mathbf{q}'_2, \ldots, \mathbf{q}'_M] \in \mathbb{R}^{M \times k}$ can be obtained for features $\mathbf{Z}$ and $\mathbf{Z}'$ assigned to each prototype. For cluster similarity matrices $\mathbf{Q}$ and $\mathbf{Q}'$, their row vectors $\mathbf{q}_i$ and $\mathbf{q}'_i$ denote the probability of assigning features $\mathbf{z}_i$ and $\mathbf{z}'_i$ of the spectral graph with its augmented view to each prototype, respectively. Each column of the clustering similarity matrices $\mathbf{Q}$ and $\mathbf{Q}'$ represents a cluster, such that $\tilde{\mathbf{q}}_k$ and $\tilde{\mathbf{q}}'_k$ denote the kth column of the clustering similarity matrices $\mathbf{Q}$ and $\mathbf{Q}'$, respectively. We want the spectral graph G_i and its augmented view G'_i to be clustered into the same cluster since the essential semantic information of the spectral graph G_i and its augmented view G'_i should be the same. This allows us to consider clusters $\tilde{\mathbf{q}}_k$ and $\tilde{\mathbf{q}}'_k$ as a pair of positive sample pairs, and the similarity between the cluster pairs can be measured by the cosine distance, that is,

$$s\left(\tilde{\mathbf{q}}_k, \tilde{\mathbf{q}}'_k\right) = \left(\tilde{\mathbf{q}}_k\right)\left(\tilde{\mathbf{q}}'_k\right)^{\mathrm{T}} / \|\tilde{\mathbf{q}}_k\|_2 \|\tilde{\mathbf{q}}'_k\|_2. \tag{8}$$

For maximizing the similarity of semantically similar spectral graph clustering clusters and keeping other semantic spectral graph clustering clusters away from them, the loss for a given spectral graph G_i is defined as

$$l_i = -\log \frac{\exp\left(s\left(\tilde{\mathbf{q}}_i, \tilde{\mathbf{q}}'_i\right)/\tau_p\right)}{\sum_{j=1}^{k}\left[\exp\left(s\left(\tilde{\mathbf{q}}_i, \tilde{\mathbf{q}}'_i\right)/\tau_p\right) + \exp\left(s\left(\tilde{\mathbf{q}}_i, \tilde{\mathbf{q}}'_j\right)/\tau_p\right)\right]} \tag{9}$$

where τ_p is the temperature parameter that controls softness, to learn the similarity between all pairs of positive samples at the level of spectral graph clustering clusters, the clustering contrast loss is computed on each cluster, which can be formalized as follows:

$$L_{\mathrm{contrast}} = \frac{1}{2k}\sum_{i=1}^{k}\left(l_i + l'_i\right). \tag{10}$$

To maintain the consistency of the $\mathbf{Z}$ and $\mathbf{Z}'$ cluster assignments, we add a regularization term for the loss of cluster consistency, $L_{\mathrm{consistency}}$, by minimizing the average cross-entropy of the $\mathbf{Z}$ and $\mathbf{Z}'$ cluster assignments, that is,

$$L_{\mathrm{consistency}} = -\frac{1}{M}\sum_{i=1}^{M}\left[\sum_{k}\mathbf{q}_i^{'(k)}\log\mathbf{q}_i^{(k)} + \sum_{k}\mathbf{q}_i^{(k)}\log\mathbf{q}_i^{'(k)}\right] \tag{11}$$

where $\mathbf{q}_i^{(k)}$ and $\mathbf{q}_i^{'(k)}$ are the probabilities that $\mathbf{z}_i$ and $\mathbf{z}'_i$ belong to the cluster centered on the kth prototype, respectively. Specifically, $\mathbf{q}_i^{(k)}(\mathbf{q}_i^{'(k)})$ is the probability that $\mathbf{z}_i(\mathbf{z}'_i)$ is assigned to the cluster centered on prototype k obtained by calculating the similarity of $\mathbf{z}_i(\mathbf{z}'_i)$ to prototype k, that is,

$$\mathbf{q}_i^{(k)} = \frac{\exp\left(\frac{1}{\tau_c}\mathbf{z}_i\mathbf{p}_k{}^{\mathrm{T}}\right)}{\sum_{k'}\exp\left(\frac{1}{\tau_c}\mathbf{z}_i\mathbf{p}_{k'}{}^{\mathrm{T}}\right)} \tag{12}$$

where τ_c is the temperature parameter that controls the consistency of clustering.

Finally, the model can discriminate between spectral similarity and dissimilarity by simultaneously optimizing the cluster contrast loss and the cluster consistency loss. The overall objective function consists of the clustering contrast loss and the clustering consistency loss, which is

$$L = L_{\mathrm{contrast}} + L_{\mathrm{consistency}}. \tag{13}$$

D. Target Detection

In the target detection stage, each pixel spectrum in the HSI to be detected and the prior target spectrum are first constructed into spectral graphs according to the designed SGC method. Then, the SGT, which has acquired spectral discrimination ability after spectral discriminative learning through the spectral graph contrastive clustering assignment, is used to extract the representations of the prior target spectral graph G^* and the spectral graph G_i of the pixel to be detected, respectively. Finally, the target detection result $\mathbf{T} = [t_1, t_2, \ldots, t_{H \times W}]$ is obtained based on the cosine distance between the representation $\mathbf{h}_i$ of the pixel spectral graph to be detected and the representation $\mathbf{h}^*$ of the prior target spectral graph. This process can be represented as

$$t_i = \cos\left(\mathbf{h}_i, \mathbf{h}^*\right) \tag{14}$$

where $\mathbf{h}_i = f_\theta(G_i)$, $\mathbf{h}^* = f_\theta(G^*)$, and t_i is the similarity of the pixel spectrum to be detected relative to the prior target spectrum.

The full process of the SGCCA is summarized in Algorithm 1.

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Algorithm 1 SGCCA

Input: HSI $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_{H\times W}] \in \mathbb{R}^{(H\times W)\times B}$; SGT $f_\theta(\cdot)$; Projection Head $g(\cdot)$; Learnable Prototypes $\mathbf{P} = [\mathbf{p}_1, \mathbf{p}_2, \ldots, \mathbf{p}_k] \in \mathbb{R}^{k\times d}$.

// Spectral Graph Construction

for $\mathbf{x}_i$ in $\mathbf{X}$ **do**
  split $\mathbf{x}_i$ into n spectral patches $\{\mathbf{x}_i^1, \mathbf{x}_i^2, \cdots \mathbf{x}_i^n\}$.
  each spectral patch is considered as a node in the spectral graph, and the spectral patch serves as the feature of the corresponding node
  connect the edge between the nodes corresponding to the two spectral patches whose positions are adjacent in the original pixel spectrum
  obtain the spectral graph G_i corresponding to the spectrum $\mathbf{x}_i$
end for

// Spectral Discriminability Learning

for *epoch* $= 1$ in *epochs* **do**
  sample a mini-batch $\{G_i\}_{i=1}^M$ from G
  for G_i in $\{G_i\}_{i=1}^M$ **do**
    randomly performing node dropping, edge perturbation and masking nodes
    obtain a spectral graph data augmented sample G_i'
  end for
  extract original spectral graph and augmented spectral graph representations by $\mathbf{y}_i = f_\theta(G_i)$, $\mathbf{y}_i' = f_\theta(G_i')$ $\mathbf{z}_i = g(\mathrm{y}_i)$, $\mathbf{z}_i' = g(\mathrm{y}_i')$
  compute the probability that $\mathbf{z}_i(\mathbf{z}_i')$ is assigned to the cluster centered on prototype k by Eq. 12
  compute the spectral graph clustering contrast loss $L_{contrast}$ by Eqs. 8-10
  compute the clustering consistency loss $L_{consistency}$ by Eq. 11
  compute the overall loss L by Eq. 13
  update $f_\theta(\cdot)$, $g(\cdot)$ through gradient descent to minimize L
end for

// Target Detection

for G_i in $\{G_i\}_{i=1}^{H\times W}$ **do**
  extract the features of the spectral graph G_i of the pixel to be detected through $\mathbf{h}_i = f_\theta(G_i)$
  extract the features of the prior target spectrum G^* by $\mathbf{h}^* = f_\theta(G^*)$
  compute the similarity between $\mathbf{h}_i$ and $\mathbf{h}^*$ via the cosine distance
end for

Output: Target detection results.

IV. Experimental Results and Analysis

A. Dataset Description

1) San Diego Dataset: An airborne visible/infrared imaging spectrometer (AVIRIS) collected the San Diego dataset [43] at San Diego airport, CA, USA. It consists of 224 bands with wavelengths ranging from 370 to 2510 nm. Bands 1–6, 33–35, 97, 107–113, 153–166, and 221–224 were removed by the low signal-to-noise ratio and water absorption, leaving 189 bands for HTD. The whole image has 400 × 400 pixels. The spatial resolution is 3.5 m, and the spectral resolution is 10 nm. In the experiment, two scene regions of size 120 × 120 and 100 × 100, named San Diego A and San Diego B, are intercepted from the upper left corner and the center of the San Diego dataset, respectively. The plane pixels in the San Diego A and San Diego B scenes are targets for HTD and contain 58 and 134 target pixels, respectively. The pseudo-color images of San Diego A and San Diego B with ground truth are shown in Figs. 4(a) and (b) and 5(a) and (b), respectively.

2) Urban Dataset: The urban dataset [60] was captured by AVIRIS sensors off the coast of Texas, USA, with a spatial resolution of 17.2 m per pixel. The Urban dataset has 100 × 100 pixels, and after removing the low signal-to-noise band, the remaining 204 bands. The detected target has a total of 67 pixels. Fig. 6(a) and (b) shows the pseudo-color image and ground truth.

3) MUUFL Gulfport Dataset: The MUUFL Gulfport dataset [61] was collected in November 2010 at the University of Southern Mississippi Gulf Park campus in Long Beach, Mississippi. The size of the original MUUFL Gulfport data scene is 325 × 337 pixels and consists of 72 bands. The first and last four bands were removed due to noise, yielding a new HSI with 64 bands. The lower right corner of the original HSI contains invalid regions, so only the first 220 columns are used for ground-truth mapping. The cropped HSI size was 325 × 220 × 64. A total of 269 clothing panel pixels in the scene were considered targets for HTD. Fig. 7(a) and (b) shows the pseudo-color images of the MUUFL Gulfport dataset with ground truth.

B. Experimental Setup

1) Comparison Methods: The experiment uses seven state-of-the-art HTD methods to compare with SGCCA. The seven comparison methods include two classical HTD methods: CEM [12] and OSP [15]. Two representation-based methods: CSCR [22] and DM-BDL [23]. Three deep-learning-based methods: MLSN [36], a detector based on an unconstrained linear mixture model and a hierarchical denoising autoencoder (denoted as ULMMDL) [62], and SCLHTD [53]. Notably, the proposed and the comparison methods both use the same a priori target spectrum for HTD.

2) Evaluation Criterion: For evaluating the HTD performance of SGCCA against seven state-of-the-art comparative methods, the 3-D receiver operating characteristic curve (3D ROC) is used to measure the detector performance. The 3D ROC curve can be considered as a function of the probability of detection, P_D, the probability of false alarm, P_F, and the threshold, τ [63]. The following procedure can calculate the P_D and P_F:

$$P_D(\tau) = \frac{n_{\mathrm{TP},\tau}}{n_{\mathrm{TP},\tau} + n_{\mathrm{FN},\tau}}$$
$$P_F(\tau) = \frac{n_{\mathrm{FP},\tau}}{n_{\mathrm{FP},\tau} + n_{\mathrm{TN},\tau}} \quad (15)$$

where $n_{\mathrm{TP},\tau}$ denotes the number of pixels that are correctly detected as targets at a given threshold τ, $n_{\mathrm{FN},\tau}$ represents the number of pixels that incorrectly detect targets as backgrounds at a given threshold τ, $n_{\mathrm{FP},\tau}$ denotes the number of pixels that

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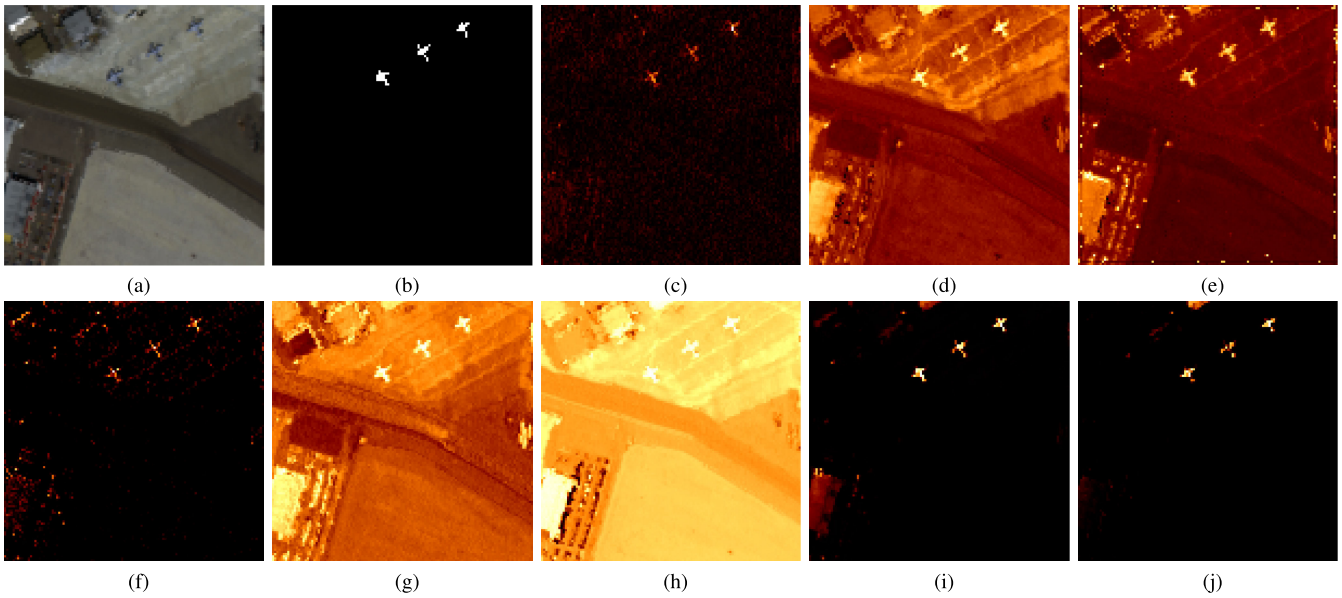


Fig. 4. Detection maps of different methods for San Diego A. (a) Pseudo-color image. (b) Ground truth. (c) CEM. (d) OSP. (e) CSCR. (f) DM-BDL. (g) MLSN. (h) ULMMDL. (i) SCLHTD. (j) SGCCA.

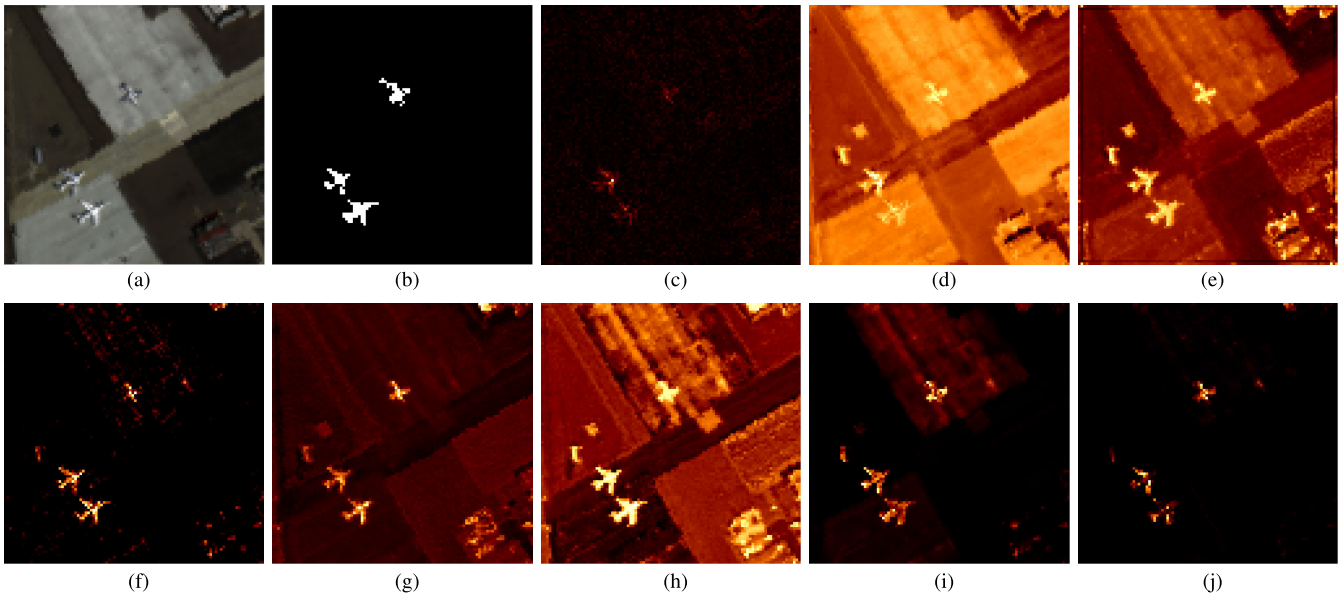


Fig. 5. Detection maps of different methods for MUUFL Gulfport. (a) Pseudo-color image. (b) Ground truth. (c) CEM. (d) OSP. (e) CSCR. (f) DM-BDL. (g) MLSN. (h) ULMMDL. (i) SCLHTD. (j) SGCCA.

incorrectly detect backgrounds as targets at a given threshold τ, and $n_{\mathrm{TN},\tau}$ denotes the number of pixels that are correctly detected as backgrounds at a given threshold τ. The 2D ROC curve of (P_D, P_F) and the 2D ROC curve of (P_F, τ) can be obtained from the 3D ROC curve. The 2D ROC curve of (P_D, P_F) can evaluate the detector effectiveness and the 2D ROC curve of (P_F, τ) can evaluate the background suppression ability of the detector. The closer the 2D ROC curve of (P_D, P_F) is to the upper left corner of the coordinate axis and the closer the 2D ROC curve of (P_F, τ) is to the lower left corner of the coordinate axis, it indicates that the detector has a better performance [64].

For the quantitative analysis of the detector, the area under the curve (AUC) of the 2-D ROC curves of (P_D, P_F) and (P_F, τ), $\mathrm{AUC}_{(P_D,P_F)}$ and $\mathrm{AUC}_{(P_F,\tau)}$, are used as quantitative indicators of the target detection and background suppression performance of the detector. The closer the $\mathrm{AUC}_{(P_D,P_F)}$ is to 1, and the closer the $\mathrm{AUC}_{(P_F,\tau)}$ is to 0, the better the target detection and background suppression performance of the detector.

3) Implementation Details: The implementation of SGCCA consists of three parts: SGC, SGDA, and spectral discriminative learning through spectral graph contrast clustering assignment. In constructing the spectral graph for the Sandiego A and Sandiego B HSIs with 189 bands, the spectral graph is constructed with seven nodes. Each node is characterized by averaging the pixel spectra into seven spectral blocks of dimension 27. The spectral graph was constructed

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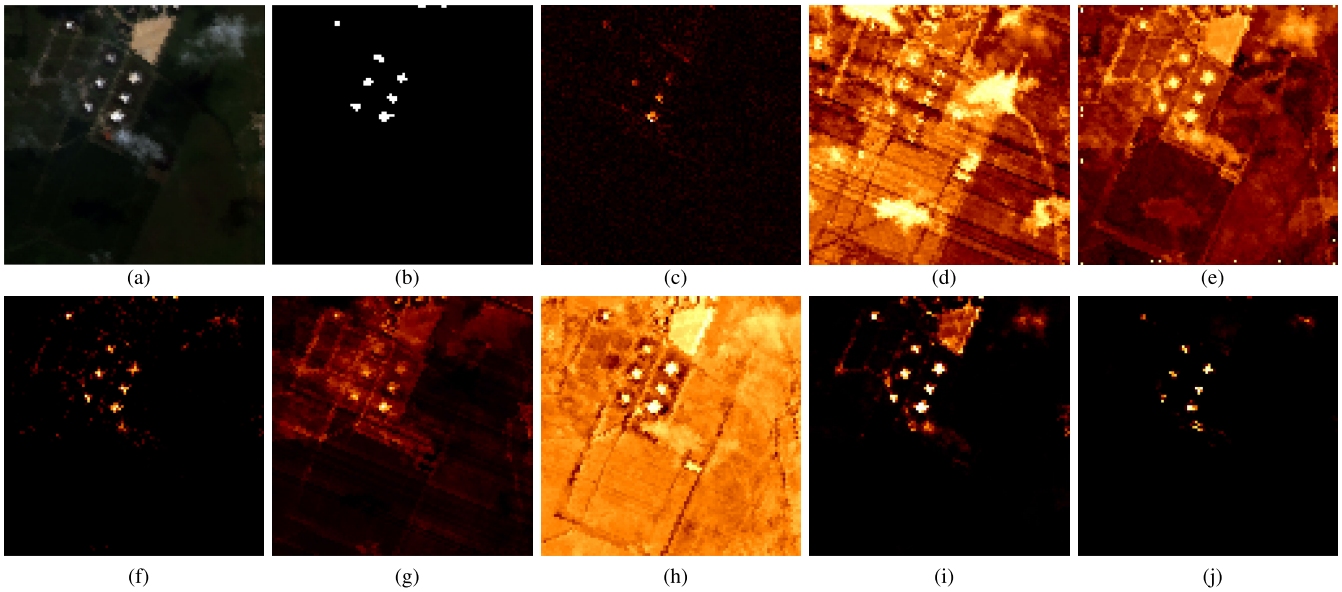


Fig. 6. Detection maps of different methods for Urban. (a) Pseudo-color image. (b) Ground truth. (c) CEM. (d) OSP. (e) CSCR. (f) DM-BDL. (g) MLSN. (h) ULMMDL. (i) SCLHTD. (j) SGCCA.

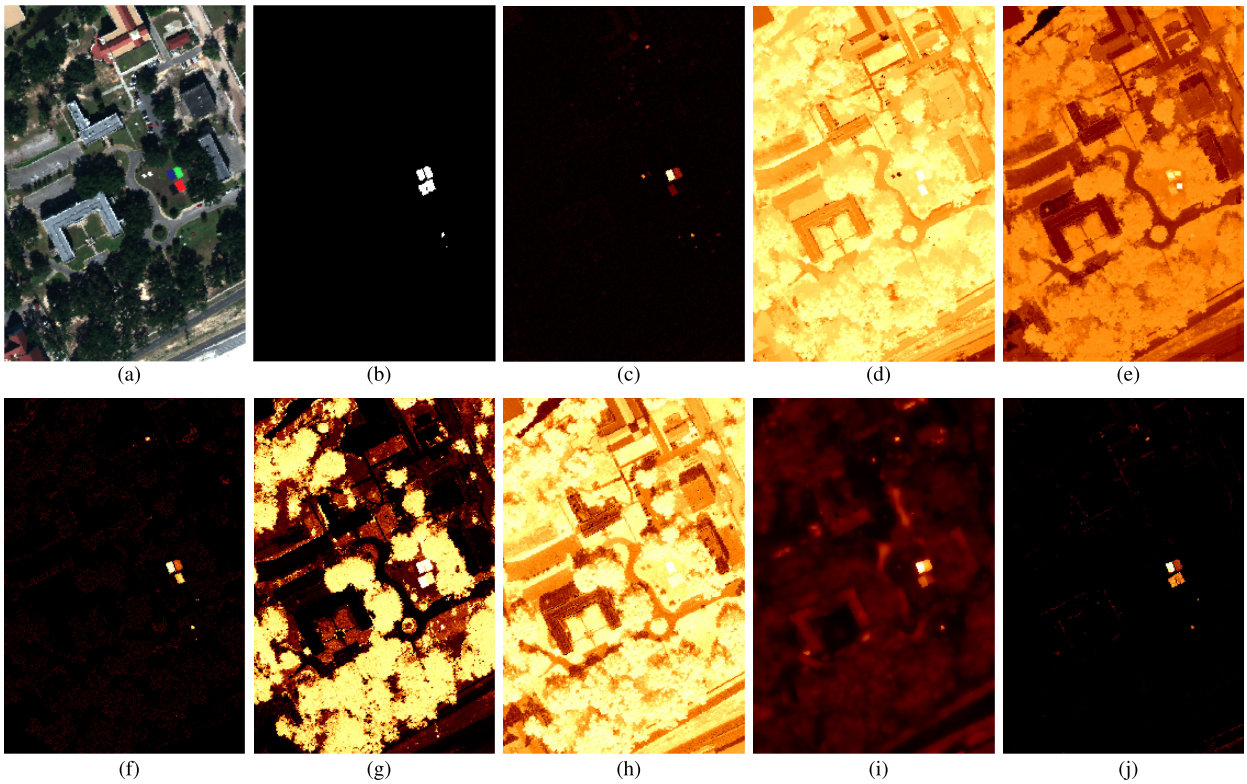


Fig. 7. Detection maps of different methods for MUUFL Gulfport. (a) Pseudo-color image. (b) Ground truth. (c) CEM. (d) OSP. (e) CSCR. (f) DM-BDL. (g) MLSN. (h) ULMMDL. (i) SCLHTD. (j) SGCCA.

with six nodes for the Urban HSI with 204 bands. Each node is characterized by averaging the pixel spectra into six spectral blocks of dimension 34. For the MUUFL Gulfport HSI with 64 bands, a spectral graph was constructed with eight nodes, each characterized by averaging the pixel spectra into eight spectral blocks of dimension 8. The spectral graphs were constructed with as few nodes per spectral graph as possible, considering the time consumption of the detectors. When performing SGDA on the constructed spectral graphs, for each spectral graph, one of the nodes dropping, edge perturbation, and masking nodes is randomly applied to produce multiple views of the spectral graph. For spectral discriminative learning via spectral graph contrast clustering assignment, the training epoch, learning rate, temperature coefficient, and number of prototypes k were set to 10, 0.001, 0.07, and 10 for the datasets used in the experiment. The SGT used for

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extracting the spectral local–global spectral features consists of a three-layer GCN and a two-layer Transformer encoder whose output feature dimension is 64. Multihead self-attention in the Transformer encoder is achieved by running $h = 4$ self-attention operations called "heads" in parallel and projecting their cascaded outputs.

Regarding the parameter settings for comparison methods, the two classical hyperspectral target detectors, CEM and OSP, do not require the setting of hyperparameters. For representation-based HTD methods, the regularization parameters λ_1 and λ_2 in CSCR are set to 0.1 and 0.01 for the datasets used in the experiments. The external and internal windows $(\omega_{\text{out}}, \omega_{\text{in}})$ are set to (7, 3), (7, 5), (9, 5), and (35, 17) for the Sandiego A, Sandiego B, Urban, and MUUFL Gulfport datasets, respectively. In DM-BDL, the decay parameter σ, learning parameter γ, and the number of atoms m is set to 0.982, 20, and 20 for all four datasets. For HTD methods based on deep learning, the inner learning rate, outer learning rate, margin, and iteration number in MLSN are set to 1e-2, 1e-3, 1, and 60 000, respectively. The learning rate and the number of hidden neurons in ULMMDL are set to 0.001 and 20 for all four datasets. The number of segmented superpixels in ULMMDL is set to 50, 50, 1000, and 50 for the Sandiego A, Sandiego B, Urban, and MUUFL Gulfport datasets. The learning rate, epoch, temperature parameter, and weight parameter β in SCLHTD are set to 0.05, 100, 0.1, and 0.5 for all four datasets.

4) Experimental Environment: The experimental hardware environment comprises an Intel Core i7-10875h 8-core CPU processor and an NVIDIA Geforce RTX 2080 GPU with 8 GB of RAM. Two classical HTD methods and two representation-based HTD methods are implemented in MATLAB R2022b, and the deep-learning-based comparison method ULMMDL is implemented using Python 3.6 and TensorFlow 1.80. The proposed SGCCA with deep learning-based comparison methods MLSN and SCLHTD is implemented using Python 3.8.3 and PyTorch 1.60.

C. Results and Discussion

1) Visual Evaluation: Figs. 4–7(c)–(j) show the detection result maps of SGCCA versus the seven comparison methods on the San Diego A, San Diego B, Urban, and MUUFL Gulfport datasets. Visually and subjectively, from the detection result maps, OSP, CSCR, MLSN, and ULMMDL can detect the target well. However, the background is not well suppressed, especially when the background of the detection scene is complex. It is not easy to recognize the target visually. DM-BDL, SCLHTD, and SGCCA have good background suppression compared to the other comparison methods, but SGCCA has better background suppression in the detection result graph. Overall, the detection results of SGCCA are closer to the ground-truth distribution of the target and show excellent detection performance than those of the comparison methods.

2) Quantitative Comparisons and Analysis: From a visual perspective, we can only roughly evaluate the target detection and background suppression performance of the detectors. To quantitatively compare the target detection and background suppression capabilities of the detector, Figs. 8–11 first show the 3D ROC and 2D ROC curves of (P_D, P_F),and 2D ROC curves of (P_F, τ) corresponding to the detection results of SGCCA and seven comparative methods on the San Diego A, San Diego B, Urban, and MUUFL Gulfport datasets. Figs. 8(b)–11(b) show the 2D ROC curves of (P_D, P_F) used to evaluate the target detection performance of the detector. To demonstrate the target detection performance of the detector under low false alarm probabilities, the horizontal axis of the 2D ROC curves of (P_D, P_F) adopts logarithmic coordinates for display, from which it can be seen that SGCCA achieves 100% target detection accuracy under the condition of the lowest false alarm probability compared to the comparison methods. Figs. 8(c)–11(c) show the 2D ROC curves of (P_F, τ) used to evaluate the background suppression performance of the detector. The (P_F, τ) 2D ROC curve of SGCCA is closer to the lower left corner compared to the comparison methods, and its background suppression capability is significantly better than the seven comparison methods in the experiment, exhibiting outstanding background suppression capability, which benefits from the more substantial spectral discrimination of the SGCCA through spectral graph contrast clustering assignment.

Since ROC curves that are very close cannot be accurately distinguished visually in terms of which detector has better performance, the area under the 2D ROC curve of (P_D, P_F) and the 2D ROC curve of (P_F, τ), $\text{AUC}_{(P_D, P_F)}$, and $\text{AUC}_{(P_F, \tau)}$ are used to accurately compare the target detection and background suppression capabilities of the detectors. Table I provides the specific values of $\text{AUC}_{(P_D, P_F)}$ and $\text{AUC}_{(P_F, \tau)}$ of SGCCA and seven state-of-the-art comparative methods on the four HSI datasets. The optimal $\text{AUC}_{(P_D, P_F)}$ and $\text{AUC}_{(P_F, \tau)}$ are highlighted in bold. It can be seen from Table I that SGCCA always obtains the highest $\text{AUC}_{(P_D, P_F)}$ and low $\text{AUC}_{(P_F, \tau)}$ on the San Diego A, San Diego B, Urban, and MUUFL Gulfport datasets, validating its effectiveness and excellent target detection and background suppression capabilities in HTD. SGCCA maintains the consistency of cluster assignments between the spectral graph and its augmented views while maximizing the similarity of spectral graph clusters with similar semantics and pushing away spectral graph clusters with different semantics, which can make the model better discriminate the differences between the pixel spectra to be detected and the prior target spectrum, thereby detecting targets better and suppressing backgrounds. ULMMDL, like SGCCA, does not require labeled target and background samples for training and has also achieved competitive target detection accuracy on the datasets in the experiments. By designing a hierarchical denoising autoencoder (HDAE), ULMMDL enhances the consistency of target spectra in the hyperspectral image to be detected, which alleviates the intraclass differences of target spectra in the HSI to be detected, enabling ULMMDL to detect targets well based on prior target spectrum. However, ULMMDL performs poorly in background suppression, and it is difficult to alleviate the intraclass differences of target spectra by simple HDAE in complex background environments, which leads to decreased target detection accuracy, as evidenced by its poor performance on the MUUFL Gulfport

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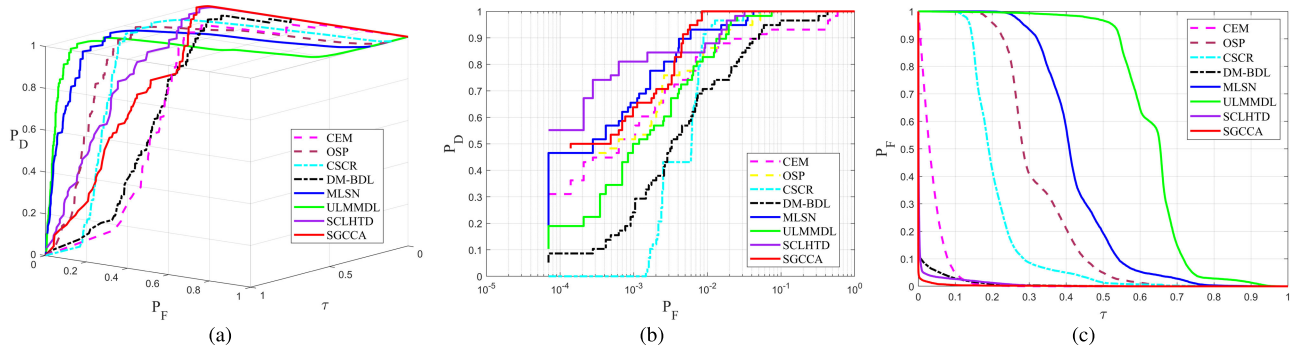


Fig. 8. Three-dimensional ROC and the corresponding 2-D ROC curves of different methods for the San Diego A dataset. (a) Three-dimensional ROC curve. (b) Two-dimensional ROC curve of (P_D, P_F). (c) Two-dimensional ROC curve of (P_F, τ).

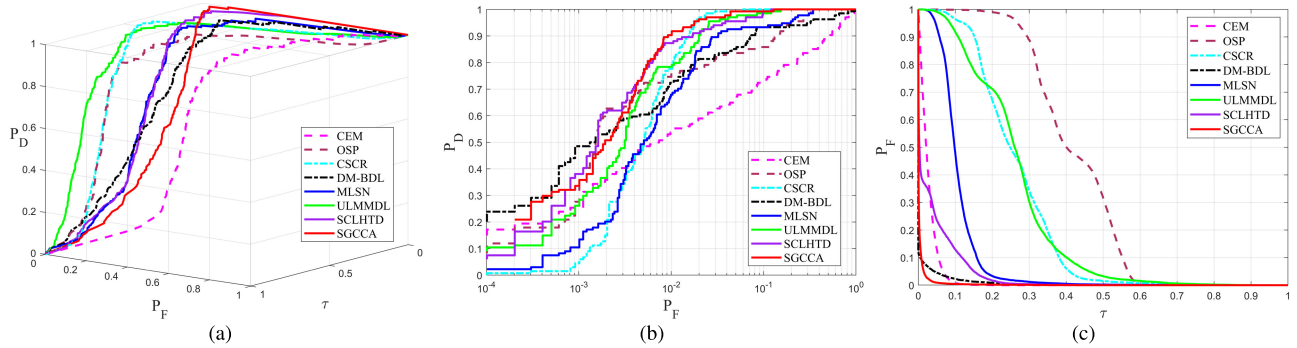


Fig. 9. Three-dimensional ROC and the corresponding 2-D ROC curves of different methods for the San Diego B dataset. (a) Three-dimensional ROC curve. (b) Two-dimensional ROC curve of (P_D, P_F). (c) Two-dimensional ROC curve of (P_F, τ).

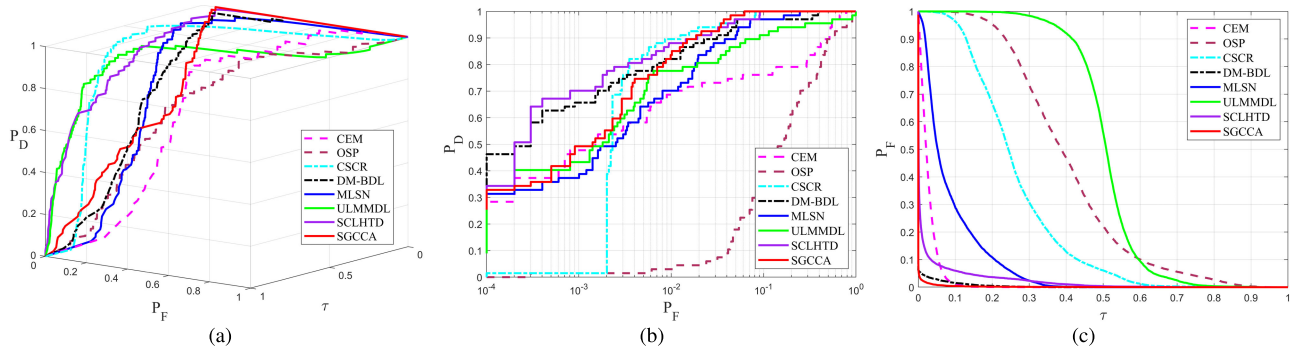


Fig. 10. Three-dimensional ROC and the corresponding 2-D ROC curves of different methods for the Urban dataset. (a) Three-dimensional ROC curve. (b) Two-dimensional ROC curve of (P_D, P_F). (c) Two-dimensional ROC curve of (P_F, τ).

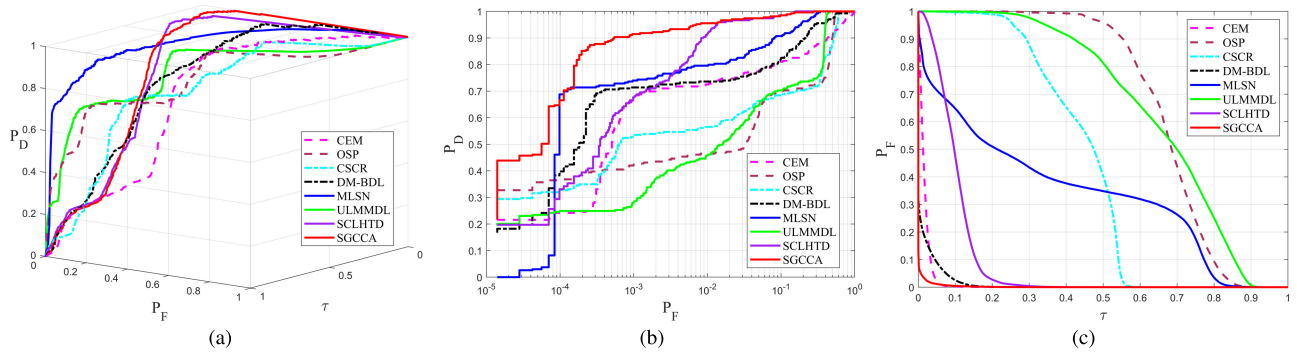


Fig. 11. Three-dimensional ROC and the corresponding 2-D ROC curves of different methods for the MUUFL Gulfport dataset. (a) Three-dimensional ROC curve. (b) Two-dimensional ROC curve of (P_D, P_F). (c) Two-dimensional ROC curve of (P_F, τ).

dataset with a complex background environment. With excellent spectral discrimination capability, SGCCA still has very competitive target detection and background suppression performance in scenes with complex backgrounds.

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TABLE I
ACCURACY COMPARISON OF DIFFERENT METHODS FOR FOUR HSI DATASETS (BOLDFACE HIGHLIGHTS THE BEST RESULT)

HSI Data Sets	Indexes	HTD Methods							
		CEM	OSP	CSCR	DM-BDL	MLSN	ULMMDL	SCLHTD	SGCCA
San Diego A	$\mathrm{AUC}_{(\mathrm{P_D},\mathrm{P_F})}$	0.96287	0.99475	0.99366	0.97586	0.99688	0.99455	0.99718	**0.99837**
	$\mathrm{AUC}_{(\mathrm{P_F},\tau)}$	0.03854	0.32051	0.21126	0.00874	0.42555	0.63472	0.00803	**0.00212**
San Diego B	$\mathrm{AUC}_{(\mathrm{P_D},\mathrm{P_F})}$	0.87253	0.96649	0.99314	0.95445	0.97638	0.99143	0.99234	**0.99488**
	$\mathrm{AUC}_{(\mathrm{P_F},\tau)}$	0.02637	0.41919	0.26017	0.00739	0.10726	0.25950	0.03340	**0.00372**
Urban	$\mathrm{AUC}_{(\mathrm{P_D},\mathrm{P_F})}$	0.89409	0.76472	0.99329	0.98404	0.98428	0.94817	0.99340	**0.99371**
	$\mathrm{AUC}_{(\mathrm{P_F},\tau)}$	0.02643	0.40561	0.26357	0.00435	0.08462	0.50771	0.02152	**0.00193**
MUUFL Gulfport	$\mathrm{AUC}_{(\mathrm{P_D},\mathrm{P_F})}$	0.90268	0.85376	0.85760	0.94445	0.97543	0.88585	0.99471	**0.99595**
	$\mathrm{AUC}_{(\mathrm{P_F},\tau)}$	0.01624	0.67801	0.43781	0.01187	0.33405	0.66187	0.09864	**0.00217**



Fig. 12. Target–background separability box plots on four HSIs. (a) San Diego A, (b) San Diego B, (c) Urban, and (d) MUUFL Gulfport.

To demonstrate the separation performance of different detectors for target and background, target–background separability boxplots are used to show the separation performance of different detectors for the target and background. Fig. 12 shows the target–background separability boxplots of the target detection results of SGCCA and seven comparative methods on four HSI datasets. In the target–background separability boxplot, boxes containing pixels with statistical distribution values of targets and backgrounds are placed, with the highest and lowest 10% of data in the target and background removed [65]. The red box indicates the distribution of targets, and the green box indicates the distribution of backgrounds. The target–background separability boxplot can reflect the separability of targets and backgrounds in

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TABLE II
IMPACT OF DIFFERENT OPTIMIZATION OBJECTIVES OF SPECTRAL GRAPH CONTRAST CLUSTERING ASSIGNMENT ON TARGET DETECTION ACCURACY

$AUC_{(P_D,P_F)}$	Sandiego A	Sandiego B	Urban	MUUFL Gulfport
$L_{contrast}$	0.90847	0.84660	0.98903	0.97576
$L_{consistency}$	0.82621	0.65261	0.88207	0.87200
$L_{contrast} + L_{consistency}$	**0.99837**	**0.99488**	**0.99371**	**0.99595**

TABLE III
TIME CONSUMPTION (IN SECONDS) FOR DIFFERENT DETECTORS ON FOUR REAL HYPERSPECTRAL DATASETS

Methods		San Diego A	San Diego B	Urban	MUUFL Gulfport
CEM		0.0139	0.0126	0.0114	0.0312
OSP		0.0774	0.0715	0.0399	0.1761
CSCR		3.3674	2.0999	2.2740	731.8416
DM-BDL		4.2168	3.4683	3.4840	19.2108
MLSN	Train	1171.483	1115.076	1109.983	1105.155
	Detect	38.4438	26.9985	31.8362	165.3926
ULMMDL	Train	73.4170	55.9274	57.0268	157.5655
	Detect	0.0781	0.0742	0.0313	0.2188
SCLHTD	Train	607.9048	463.4998	459.1462	2850.2856
	Detect	58.4382	41.9721	43.0486	276.6257
SGCCA	Train	543.2021	348.0766	353.5579	2397.0493
	Detect	10.5337	7.4410	7.9134	52.9560

the detection results and observe the distribution range of the detection values of target pixels and background pixels in the detection results. As shown in Fig. 12, SGCCA suppresses the detection values of background pixels in the detection results to a very low level in the four datasets in the experiments, demonstrating excellent background suppression capabilities, and SGCCA can also separate targets and backgrounds well. Especially on the MUUFL Gulfport dataset with a complex background in the detection scene, the targets and backgrounds in the detection results of SGCCA have good separability. At the same time, CEM, OSP, CSCR, and ULMMDL cannot separate the targets and backgrounds well. Competitive separability indicates that spectral discriminative learning through spectral graph contrast clustering assignment can enable the model to learn the capability of spectral discrimination effectively.

D. Ablation Studies

To investigate the effectiveness of the proposed spectral graph contrast clustering assignment method for the HTD task, we conducted an ablation study on the optimization objective of graph contrast clustering assignment. Spectral graph contrast clustering assignment maximizes the similarity between two spectral graph clusters with similar semantics while pushing away spectral graph clusters with different semantics while maintaining the consistency of cluster assignments between the spectral graph and its augmented graph so that the model learns the capability of discriminating spectral differences. The optimization objective of spectral graph contrast clustering assignment consists of the cluster contrast loss and the cluster consistency loss. Table II shows the impact of different optimization objectives of spectral graph contrast clustering assignment on target detection accuracy. It can be seen from Table II that jointly using the cluster contrast loss and the cluster consistency loss can achieve the highest target detection accuracy on the four datasets, while using only the cluster contrast loss or the cluster consistency loss leads to unsatisfactory target detection accuracy for the detector. The experimental results of the ablation study on the loss function demonstrate the effectiveness of the proposed method.

E. Time Consumption and Model Complexity

Table III shows the time consumption (in seconds) of different detectors on the datasets in the experiments. Table IV shows the number of model parameters and computational complexity (in millions or thousands) of four deep-learning-based detectors on four HSI datasets. The number of model parameters is the total number of learnable parameters in the network, and the computational complexity is the number of floating-point operations (FLOPs) required during the forward propagation of the network. Regarding time consumption, the traditional HTD methods CEM and OSP demonstrate competitively advantageous performance. In contrast, deep-learning-based HTD methods, such as MLSN, ULMMDL, SCLHTD, and SGCCA, require longer training times. Compared with classic and representation-based detectors, although the training time consumption of deep-learning-based detectors increases due to the need to train models, they can achieve better target detection accuracy. Compared with MLSN and SCLHTD, the model training time, detection time, and

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TABLE IV
MODEL PARAMETERS AND COMPUTATIONAL COMPLEXITY (IN MILLIONS OR THOUSANDS) OF FOUR DEEP-LEARNING-BASED DETECTORS ON FOUR HSI DATASETS

Methods	Indexes	San Diego A	San Diego B	Urban	MUUFL Gulfport
MLSN	Model Parameters(M)	0.05	0.05	0.05	0.05
	FLOPs(M)	9.83	9.83	10.56	3.31
ULMMDL	Model Parameters(K)	7.77	7.77	8.38	2.64
	FLOPs(K)	15.53	15.53	16.79	5.28
SCLHTD	Model Parameters(M)	0.22	0.22	0.22	0.22
	FLOPs(M)	12.90	12.90	12.90	12.90
SGCCA	Model Parameters(M)	0.10	0.10	0.10	0.10
	FLOPs(M)	0.68	0.68	0.62	0.42

computational complexity of SGCCA are lower than them while achieving higher target detection accuracy. Although ULMMDL is superior to SGCCA in terms of time consumption, model parameterization, and computational complexity, SGCCA can achieve higher target detection accuracy and perform better in complex background conditions.

V. CONCLUSION

To capture the local and global spectral information and enable the model to detect targets and suppress backgrounds better, we propose the HTD method SGCCA that uses SGT to extract local–global spectral information from spectra and performs spectral discriminability learning through spectral graph contrast clustering assignment. Specifically, pixel spectra are first constructed as spectral graphs, and augmented graphs of spectral graphs are obtained by randomly applying node dropping, edge perturbation, and masking nodes to each spectral graph. Then, local–global features of spectral graphs and augmented graphs are extracted by SGT and clustered into k clusters using a set of prototype vectors. Second, the similarity between spectral graph clusters with similar semantics is maximized while pushing away spectral graph clusters with different semantics so that the model can discriminate spectral differences. To maintain the consistency of clustering assignments between the spectral graph and its augmented graph, we add a cluster consistency loss to help the model learn better spectral discrimination. Notably, SGCCA performs spectral discriminability learning in a self-supervised manner. At detection, the spectrally discriminative SGT learned simultaneously distinguishes the similarities and differences between the pixel spectra to be detected and the prior target spectrum to highlight targets and suppress backgrounds. Finally, experimental results on four real HSIs and ablation studies demonstrate that SGCCA can achieve excellent target detection and background suppression performance on the HTD task.

Furthermore, leveraging the significant spectral discriminative capability of SGCCA, we conducted cross-test experiments on the San Diego A and B datasets. The results demonstrate that SGCCA maintains its detection capability. The model trained on the San Diego A dataset achieved an AUC of 0.8446 on the San Diego B dataset, while the model trained on the San Diego B dataset achieved an AUC of 0.9733 on the San Diego A dataset. However, due to the small data volume of these two datasets, the models could not acquire a broader spectral discriminative capability, leading to suboptimal discriminative performance. Moreover, the extensive training time required for SGCCA hinders its real-time application, signifying a need for further optimization.

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